

AZURE + MICROSOFT FABRIC

Make technology spend explainable. Make its business value testable.

AERO connects workload decisions, cost evidence, ownership and quality so clients can evaluate what to optimize - and what is worth protecting.

Clients need more than a spend total. They need to know why work ran, who can act, and whether the output justified its cost. AERO provides a functional local decision and evidence layer; Phase 2 extends that model to a 360-degree view across Azure and Fabric.

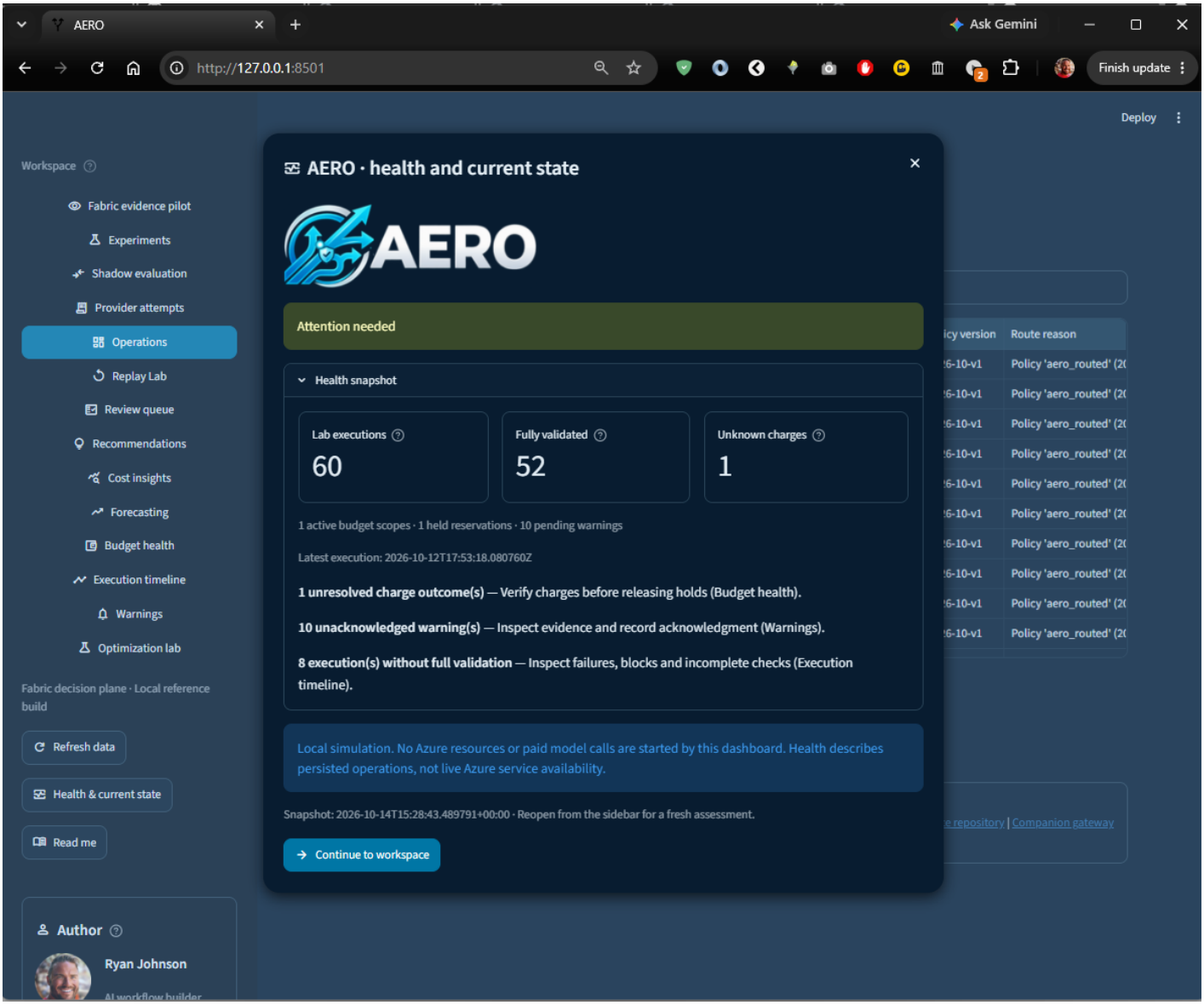
Why I started AERO. I initiated this project to address the most pressing capabilities clients seek in better end-to-end Azure solutions: visibility, practical optimization, accountable governance and measurable business outcomes.

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AERO / FULL APPLICATION SCREENSHOT

A working solution, in context.

The complete health-and-current-state screenshot shows the application, workspace navigation and the evidence that requires operator attention.



AERO Health and current state | Full screenshot

The local showcase reports 60 executions, 52 fully validated results, one unknown charge and 10 warnings. Client benefit: decision-makers see outstanding evidence and the next action in one place. Skill demonstrated: operational product design and transparent exception handling.

These are local demonstration records, not customer deployment results. No Azure resources or paid model calls are started by this dashboard. [1]

01 / COMPREHENSIVE VISIBILITY

Connect the bill to the work.

For clients, the opportunity is a common explanation of spend that finance, engineering and business owners can use together.

AERO records workload, team, application and environment alongside policy version, route reason, validation, latency and cost evidence. Immutable cost facts preserve source, currency, pricing basis and freshness. Re-imports are deduplicated; conflicting records are rejected. [1, 2]



01 Cost insights | Selected detail

Unit economics connect spend to teams, applications and validated outputs. Client benefit: owners can compare like-for-like workload costs and identify the accountable team. Skill demonstrated: financial evidence modeling and analytical product design.

A 360-DEGREE DESIGN, WITH VISIBLE COVERAGE

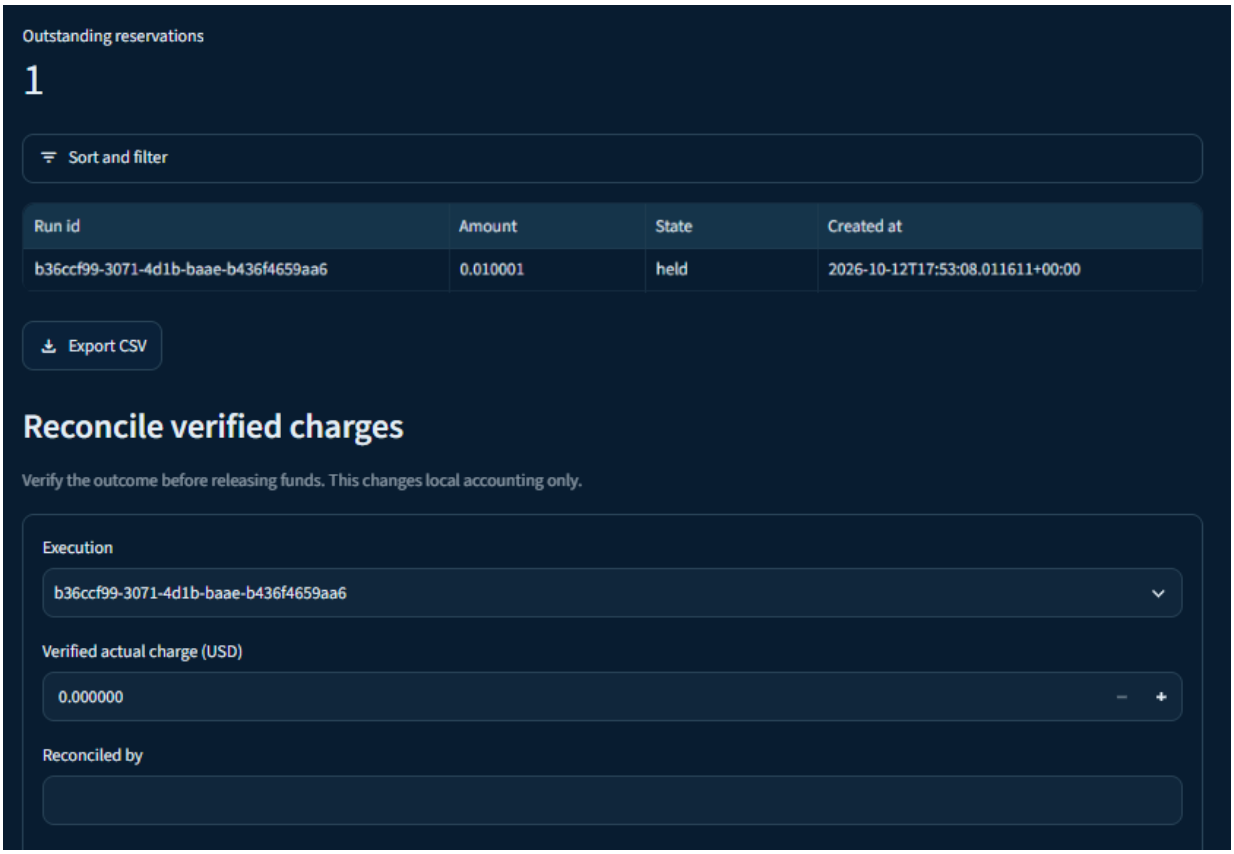
Phase 2 connects pipeline, notebook, query, refresh, OneLake and adjacent Azure evidence to one business run. Reliable identifiers anchor joins; unmatched costs and stale sources stay visible. Trace duration explains operations; billing and usage establish cost. [3]

02 / GOVERNANCE AND ACCOUNTABILITY

Decide before dispatch. Reconcile after execution.

AERO makes the budget decision part of the workflow, with durable evidence for the person responsible.

Before managed local execution, AERO reserves estimated cost atomically against matching organization, team, application and environment limits. Spend and reservations survive restarts. Uncertain charges retain a hold until an operator supplies verified evidence; they do not become available funds by default. [1, 2]



02 Budget health | Selected detail

A held reservation and explicit reconciliation form make uncertainty visible. Client benefit: repeatable spending decisions and a named reviewer. Skill demonstrated: transaction design, failure handling and operational accountability.

CLEAR CONTROL BOUNDARIES

Admission caps govern new AERO executions. They do not stop in-flight jobs or cap the whole Azure invoice. Idempotent API retries reuse the original receipt. Review takes precedence over automated routing, and recommendation approval does not deploy a policy. [1, 2]

03 / ACTIONABLE OPTIMIZATION

Make a cheaper path earn approval.

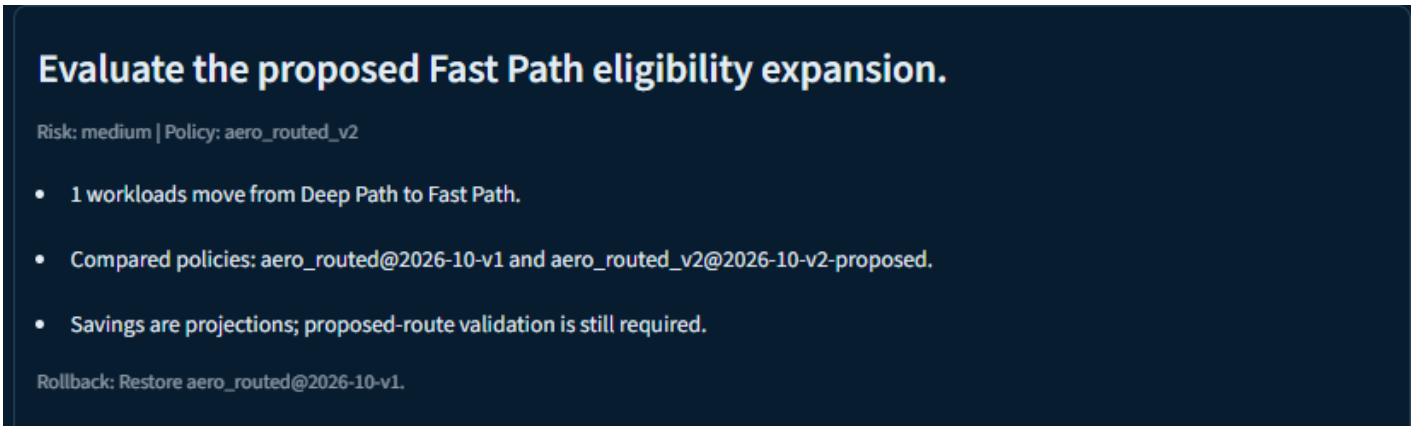
AERO moves from a cost signal to a testable change, preserving quality and keeping the owner in control.

Replay Lab compares versioned policies over the same cohort, exposing changed routes, projected cost, budget blocks and review burden. Optimization Lab then executes equivalent local transforms and checks matching outputs, elapsed time, p95 and CPU usage. Elapsed time includes AERO overhead. [1, 2]



03 Optimization Lab | Comparison detail

The sample shows matching outputs but does not satisfy every improvement criterion. Client benefit: a faster result is tested before adoption. Skill demonstrated: honest benchmarking and quality-aware optimization. Prices are illustrative; this is not measured Azure savings.



04 Governed recommendations | Decision detail

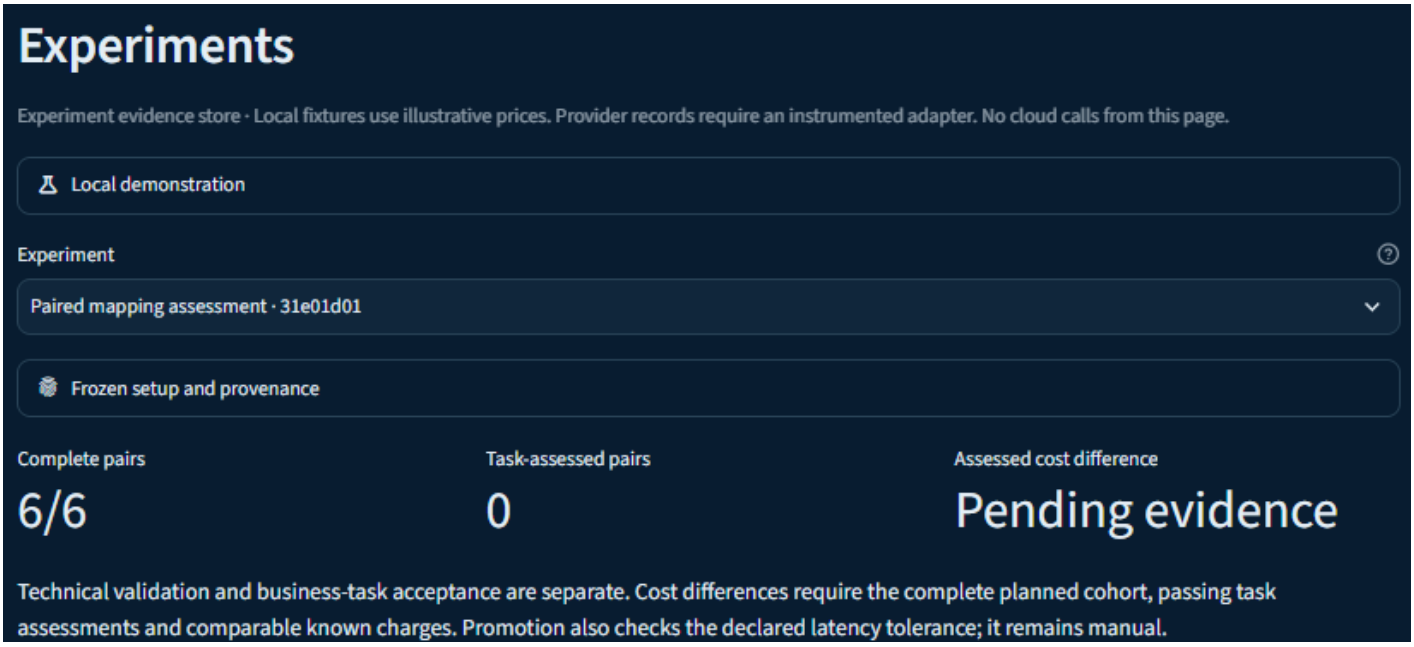
The recommendation names the policy change, risk, validation requirement and rollback version. Client benefit: an optimization can be evaluated and approved deliberately. Skill demonstrated: translating technical evidence into a reviewable change.

04 / BUSINESS-VALUE MEASUREMENT

Count accepted outcomes. Include the cost of getting there.

Lower cost is useful only when the business receives the quality, coverage and timeliness it needs.

AERO separates technical validation from business-task acceptance. Frozen experiment cohorts and setup records bind receipts to a fair baseline. A total cost difference requires a complete assessed cohort and comparable known charges; latency tolerance is declared in advance and promotion remains manual. [1]



05 Experiments | Acceptance detail

Six completed pairs remain unassessed, so the cost difference stays pending. Client benefit: completion is not mistaken for business success. Skill demonstrated: evaluation design, provenance and evidence-based commercial judgment.

ILLUSTRATIVE CLIENT SCENARIO: VALIDATED INVOICES

For a distributor, the outcome is a validated invoice record. Use structured mapping where sufficient, OCR for scans and reasoning or review for unresolved ambiguity. Measure total cost per accepted invoice, exact field agreement, review effort and turnaround. Azure execution adapters are Phase 2. [1]

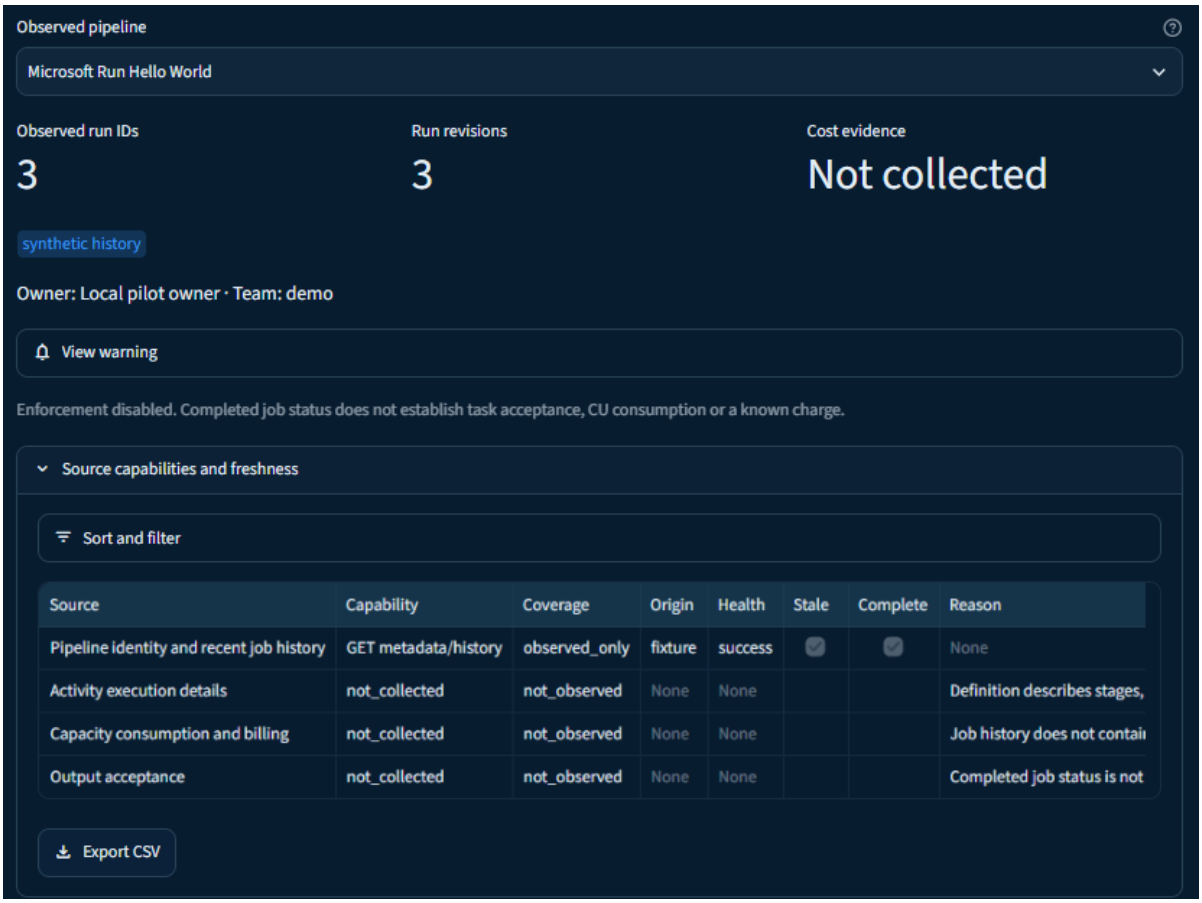
Credible value reporting: distinguish less consumption, more capacity headroom, forecast avoided capacity growth and realized cash savings. An unchanged fixed-capacity bill may stay unchanged even when allocated workload cost falls. Include failed attempts, verification, review and AERO operating overhead. [1, 3]

05 / PHASE 2 - WORK IN PROGRESS

Extend the evidence model across Azure and Fabric.

The long-term solution centers on one business execution graph: who requested work, what triggered it, which services ran and whether the outcome was useful.

The read-only collection foundation provides scoped metadata/history GETs, validated imports, deduplication and coverage/freshness reporting. The displayed history is synthetic; no real workspace has been observed. Phase 2 develops live integration, cross-service attribution and approved execution controls. [1, 3]



06 Fabric evidence pilot | Coverage detail

Pipeline history is observed while billing and output acceptance are explicitly missing. Client benefit: gaps remain visible before decisions are trusted. Skill demonstrated: integration design, observability and disciplined scope control.

What this enables for clients: explain end-to-end workload cost; investigate retries, fan-out and repeated refreshes; compare approved incremental and full processing paths; align optional work with capacity headroom and business deadlines; assign storage and monitoring opportunities to owners. Phase 2 complements Fabric native metrics and Azure Cost Management. [3]

06 / MY CONTRIBUTION



Client needs. Proven judgment.

AERO is a portfolio example of the initiative, technical depth and consulting judgment I bring to a role.

Client-led initiative	I started with questions clients need answered: what drove spend, who owns it, which changes are safe and what value the output creates.
Coherent engineering	I connect typed workload contracts, versioned policy, FastAPI, durable SQLite accounting, OpenTelemetry and an evidence-focused interface into one decision flow. [1, 2]
Commercial judgment	I distinguish projections from measured results, capacity headroom from cash savings, and technical completion from accepted business outcomes.
Delivery discipline	The documented local lab includes 231 passing tests covering concurrency, rollover, cost imports, unknown-charge holds and incorrect candidate outputs. That evidence supports the local build. [2]

This approach creates a concrete basis for discovery workshops, evidence-led optimization assessments and a governed client pilot. I bring the ability to connect engineering choices to client economics, define acceptance before automation and explain tradeoffs clearly.

PROJECT SCOPE AND COMPLETION COMMITMENT | OCTOBER 14, 2026

The local solution is fully functional. My commitment is to make the initial local solution 100% complete by Friday, October 16, 2026. AERO remains a work in progress overall; Phase 2 covers live Azure/Fabric adapters, production identity, tenant isolation and shared durable storage. Demonstration prices and billing are synthetic; Deep Path is simulated.

INDUSTRY CONTEXT

Microsoft's October 9 announcement describes structured decision scoring for routing, prioritization and workflow control. AERO applies a complementary principle: decisions need policy, budget, evidence and review. Microsoft-Decision-1 is an optional future adapter choice, not an established AERO integration. AERO is independent; no Microsoft endorsement is implied. [4]

SOURCE NOTES

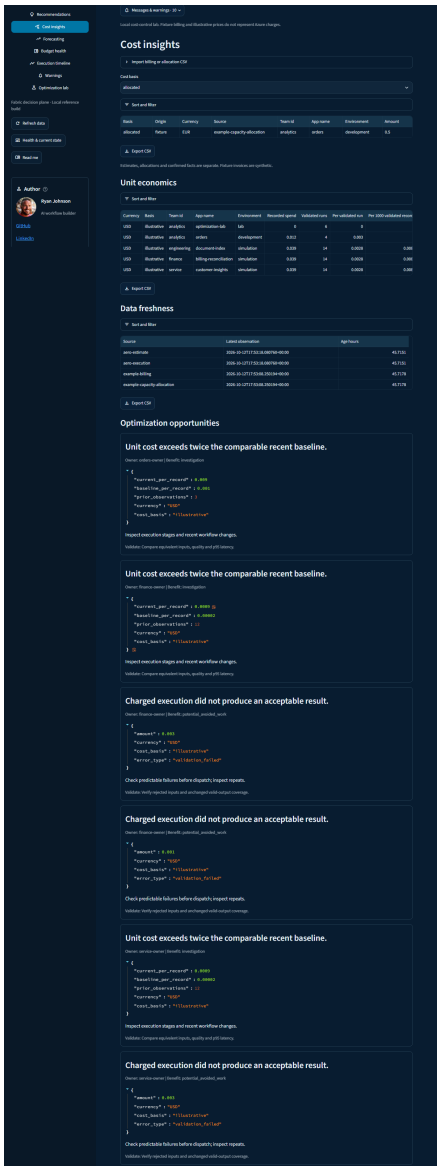
[1] AERO README - capabilities, scenarios and operating limits.
[2] Local Cost Control Lab - implementation and verification evidence.
[3] Fabric Cost Governance Vision - Phase 2 architecture and benefit definitions.
[4] Microsoft: Introducing Microsoft-Decision-1 - October 9, 2026.
Sources reviewed October 14, 2026. Screenshots: supplied local showcase, selected detail crops. Client scenarios are illustrative, not deployment or savings claims.

APPENDIX / AERO FULL SCREENSHOTS

Cost evidence and task acceptance.

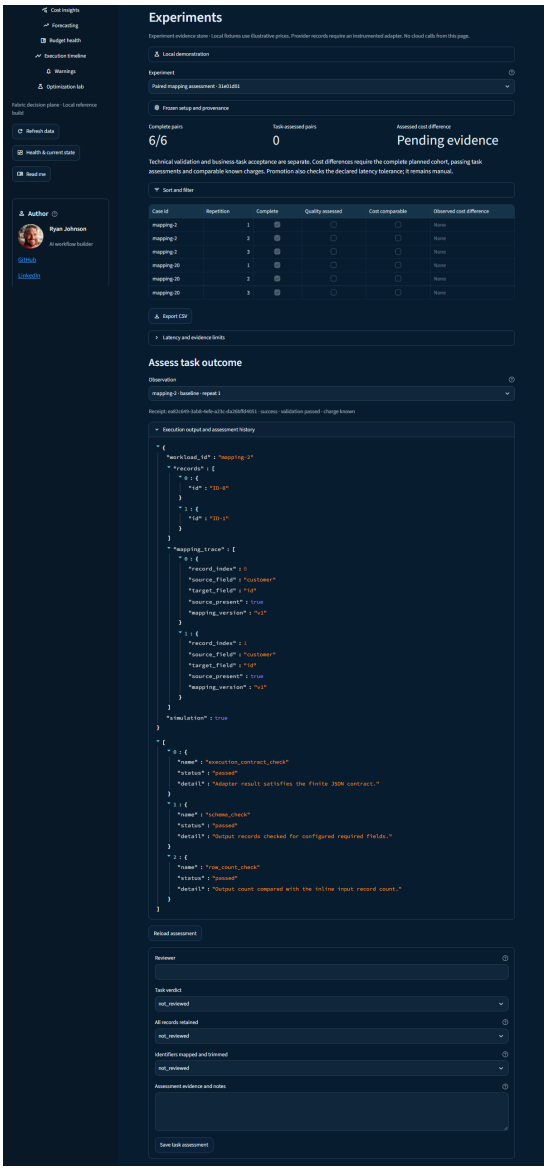
Complete supplied images, shown without cropping or distortion. Enlarge the PDF to inspect fine detail; demonstrations use local data and illustrative charges.

Cost insights



Full view behind page 3. Ownership, unit economics, freshness and evidence-backed opportunities connect cost to an accountable decision.

Experiments



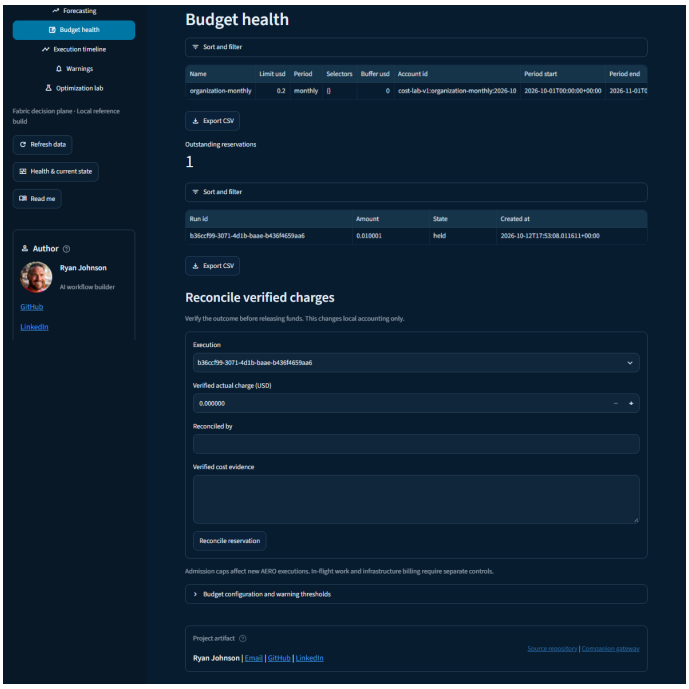
Full view behind page 6. Frozen setup, output evidence and reviewer assessments separate technical completion from business-task acceptance.

APPENDIX / AERO FULL SCREENSHOTS

Budget control and validated change.

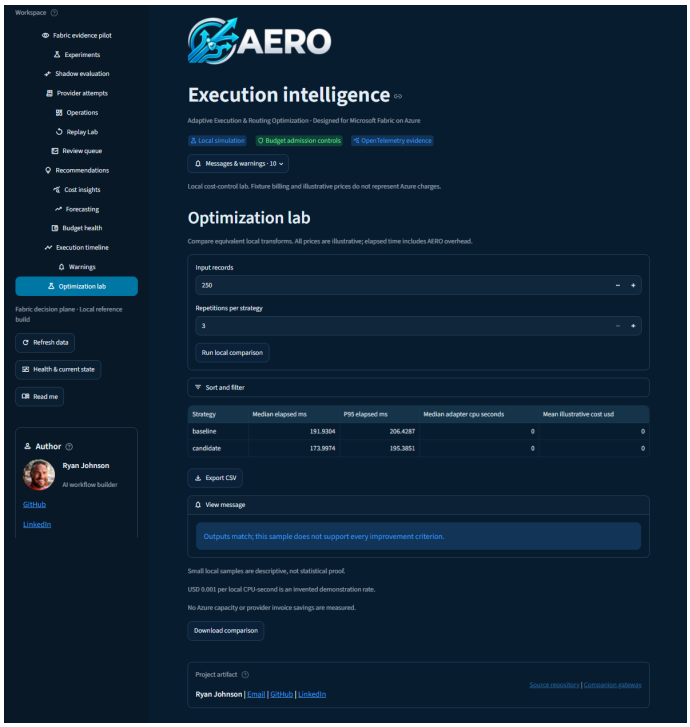
Complete supplied images, shown without cropping or distortion. Enlarge the PDF to inspect fine detail; demonstrations use local data and illustrative charges.

Budget health



Full view behind page 4. Held reservations require verified charges and a reviewer before funds are reconciled.

Optimization Lab



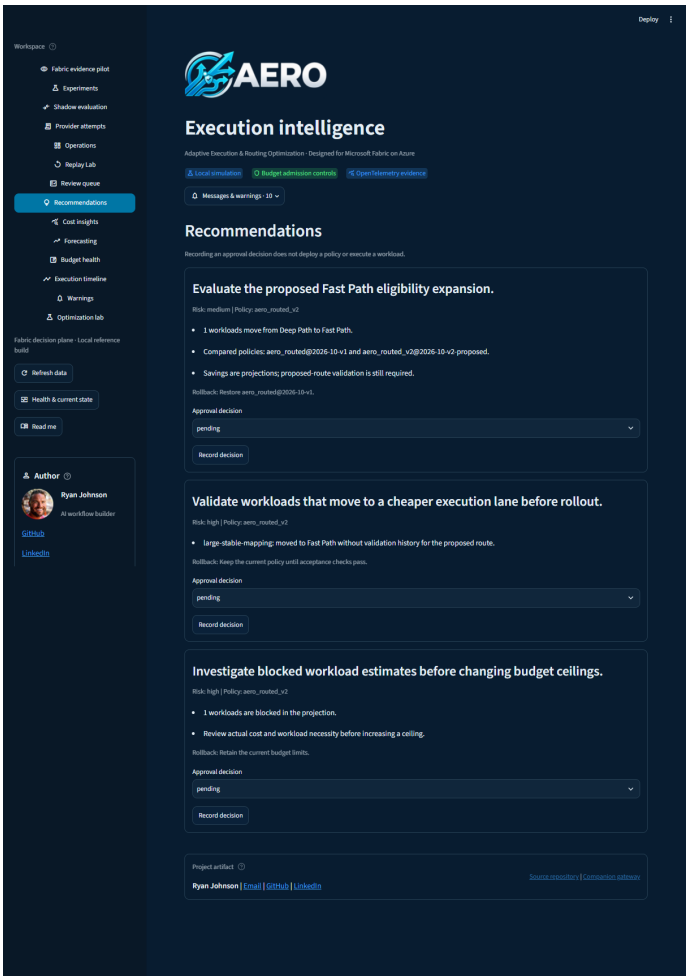
Full view behind page 5. Equivalent local transforms are compared; matching outputs alone do not establish every improvement criterion.

APPENDIX / AERO FULL SCREENSHOTS

Governed decisions and source coverage.

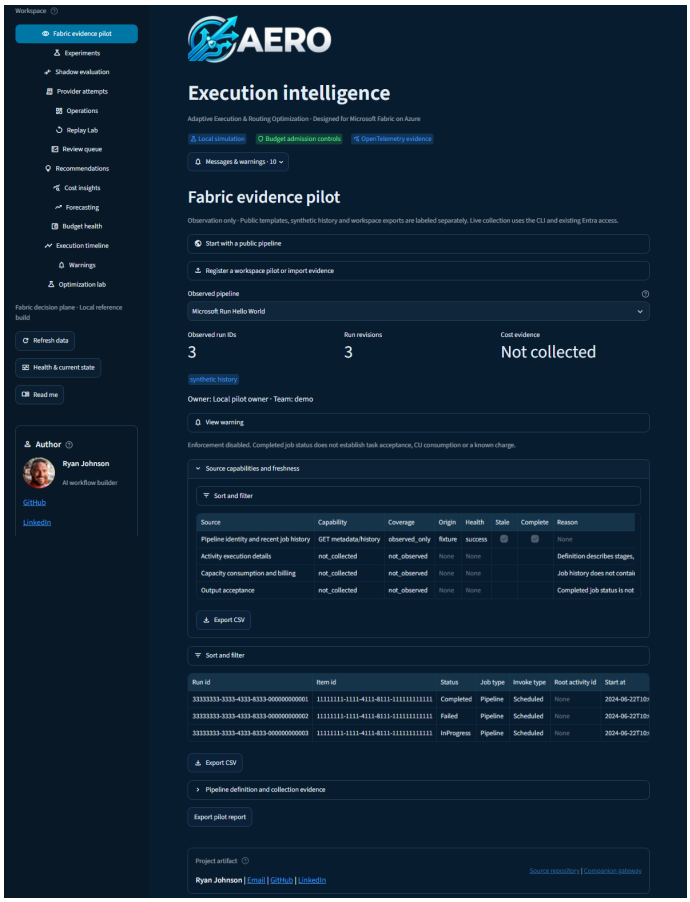
Complete supplied images, shown without cropping or distortion. Enlarge the PDF to inspect fine detail; demonstrations use local data and illustrative charges.

Recommendations



Full view behind page 5. Proposed policy changes carry risk, validation requirements, approval state and rollback notes.

Fabric evidence pilot



Full view behind page 7. Synthetic history, source capabilities and missing cost/acceptance evidence remain explicit.

OTHER PUBLISHED PROJECTS / AI COST GOVERNANCE

LLM Cost and Evaluation Gateway

A reusable execution layer that centralizes model calls, pre-call budget reservation, configurable routing, bounded retries and a SQLite usage ledger.

View project on GitHub | Source: published README and supplied screenshot paths.



Full overview screenshot. Estimated spend, a strong-only baseline and routing projections make the commercial effect of model choice visible. The figures represent a synthetic projected ledger, not realized client savings.



Full cost-drivers screenshot.
Cost by incident type and model/route mix show where an optimization assessment should begin.

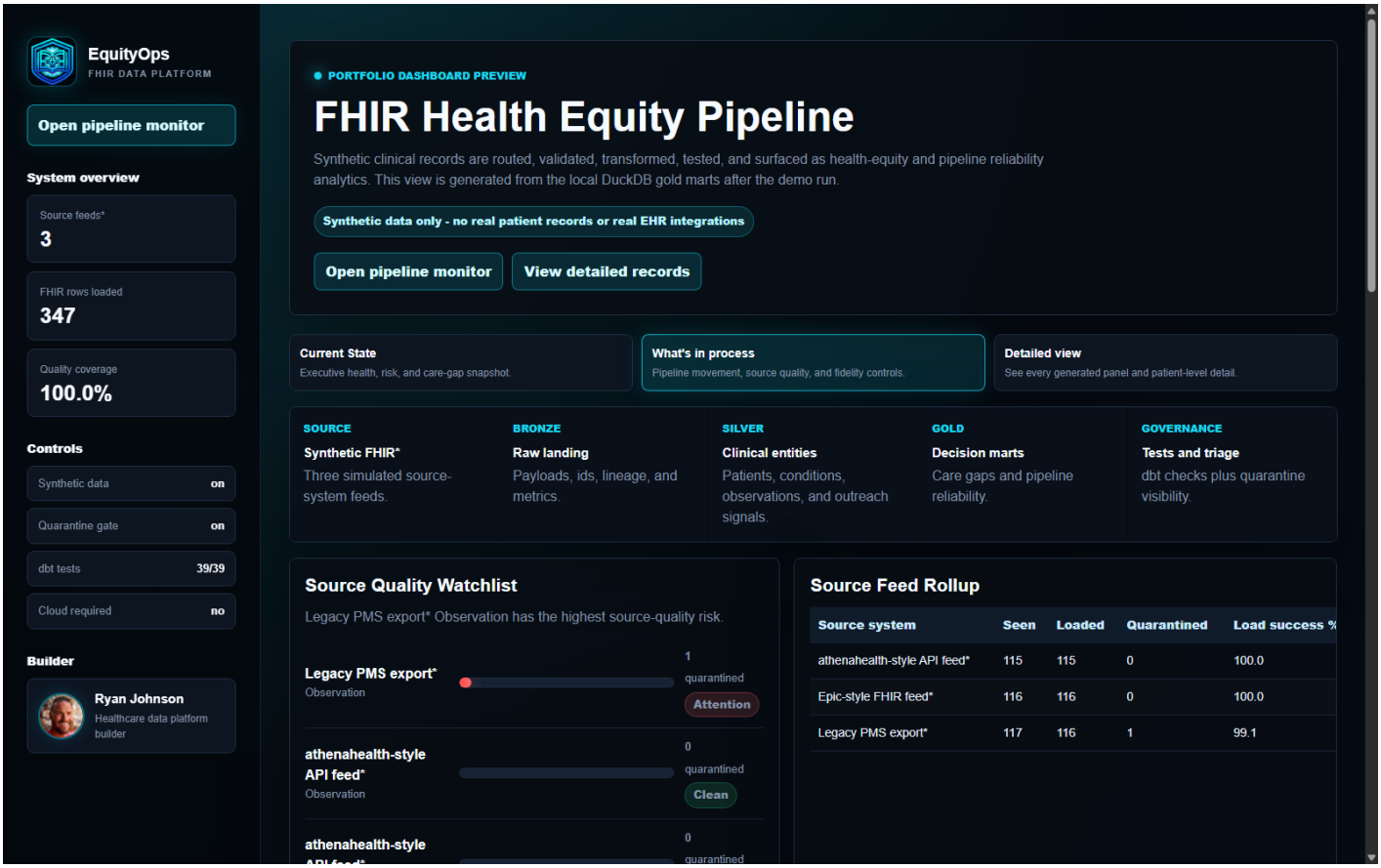
What I bring: reusable API design, cost attribution and a clear explanation of model tradeoffs. This foundation informs AERO's broader workload controls.

OTHER PUBLISHED PROJECTS / DATA ENGINEERING

FHIR Health Equity Pipeline

A local data platform that routes synthetic FHIR-shaped records into DuckDB, tested dbt transformations and stakeholder-ready care-gap and reliability marts.

View project on GitHub | Source: published README and supplied screenshot paths.



Full process-view screenshot. Source, bronze, silver, gold and governance stages appear alongside source-quality and feed rollups. The displayed vendor-style feeds are simulated; they are not real EHR integrations.

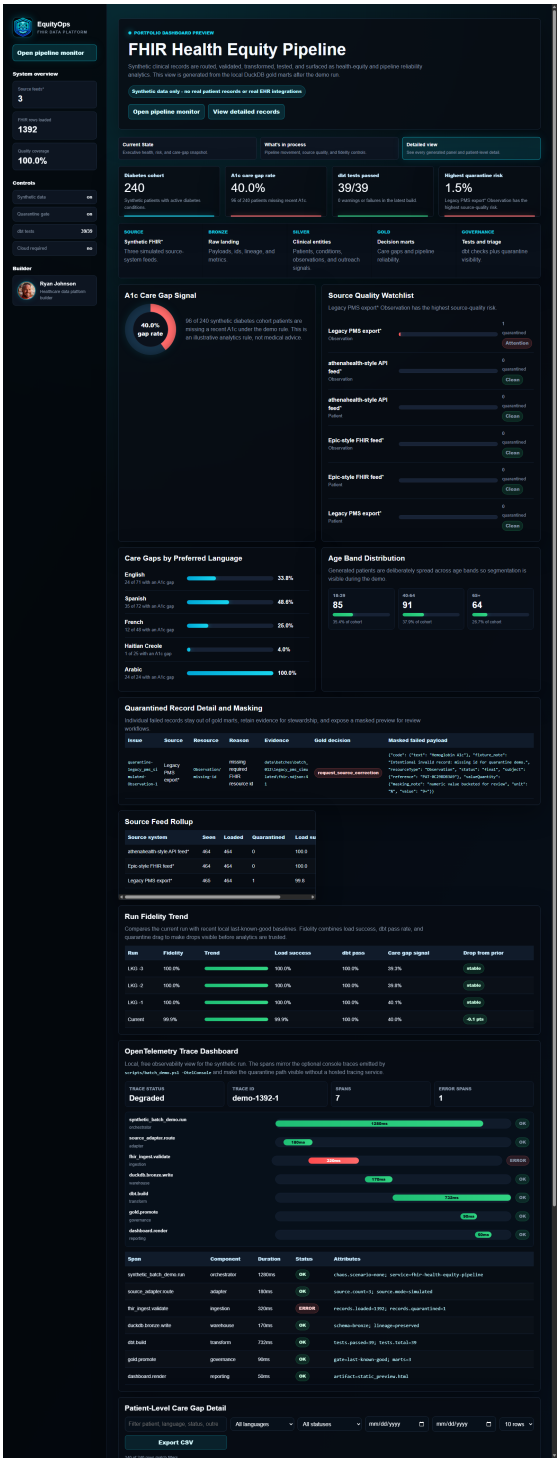
ENGINEERING PATTERNS THAT TRANSFER TO CLIENT WORK

Validation before promotion, quarantine with lineage, 39 dbt tests, quality triage and last-known-good snapshots make analytical trust inspectable. The project demonstrates how I connect data engineering, operational reliability and stakeholder questions in one usable product.

OTHER PUBLISHED PROJECTS / FULL SCREENSHOT

FHIR: from source quality to insight.

The complete detailed dashboard connects care-gap analysis, masked failed records, source rollups, fidelity, traces and patient-level inspection.



FULL DETAILED VIEW

The full supplied screenshot is preserved at its original aspect ratio. Enlarge the PDF to inspect the long dashboard.

WHAT THE CLIENT CAN EVALUATE

Care-gap segmentation links analytics to a stakeholder question. Quarantine detail and masking expose quality issues before records reach trusted outputs. Fidelity and trace panels connect the analytical result to pipeline reliability.

WHAT THIS DEMONSTRATES

I translate domain requirements into tested data models, make exceptions visible and design different levels of detail for business and technical reviewers. These patterns support AERO's emphasis on accepted outcomes and explainable execution.

EVIDENCE BOUNDARY

All records are synthetic. Care-gap metrics illustrate analytics behavior; they are not clinical findings. Masking demonstrates a design pattern and does not establish HIPAA compliance or production adoption.

Explore the FHIR project
Open my full project portfolio

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